

DATUM LINE

AI × BIO · 29 SEPT

“Do my cells look normal?”

Drop an image. **Get an answer** against every
other lab.

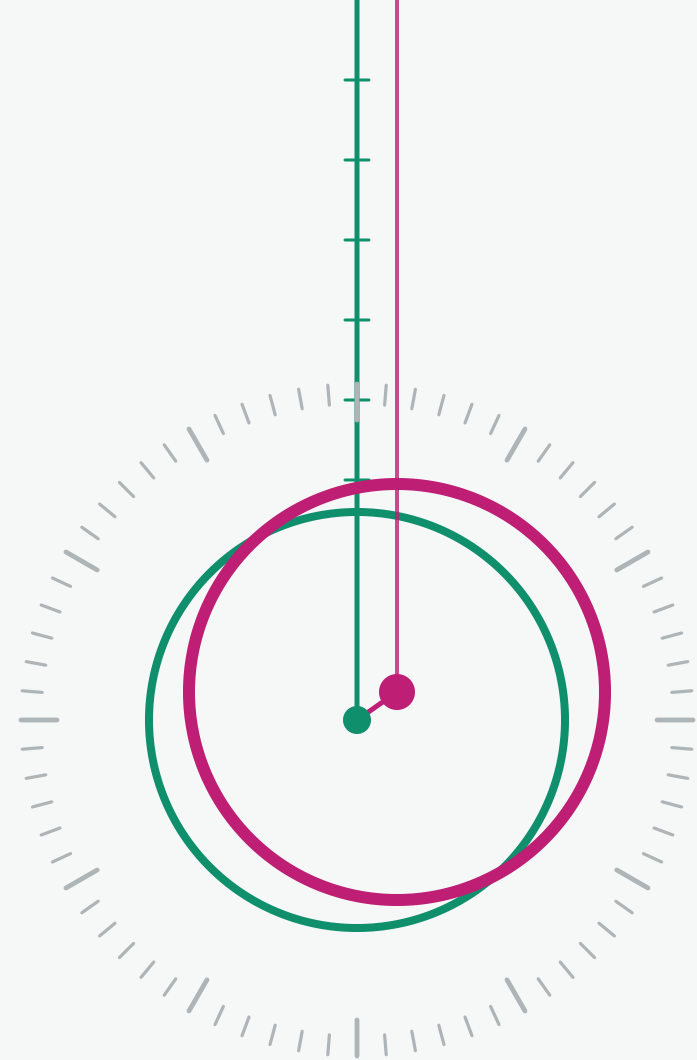


a control image, from any microscope
no compound, no target, no result

ask

Labs ask for free. The companies that buy cell data pay for the
certificate.

Kat Titterton · San Francisco



One lab, or six?



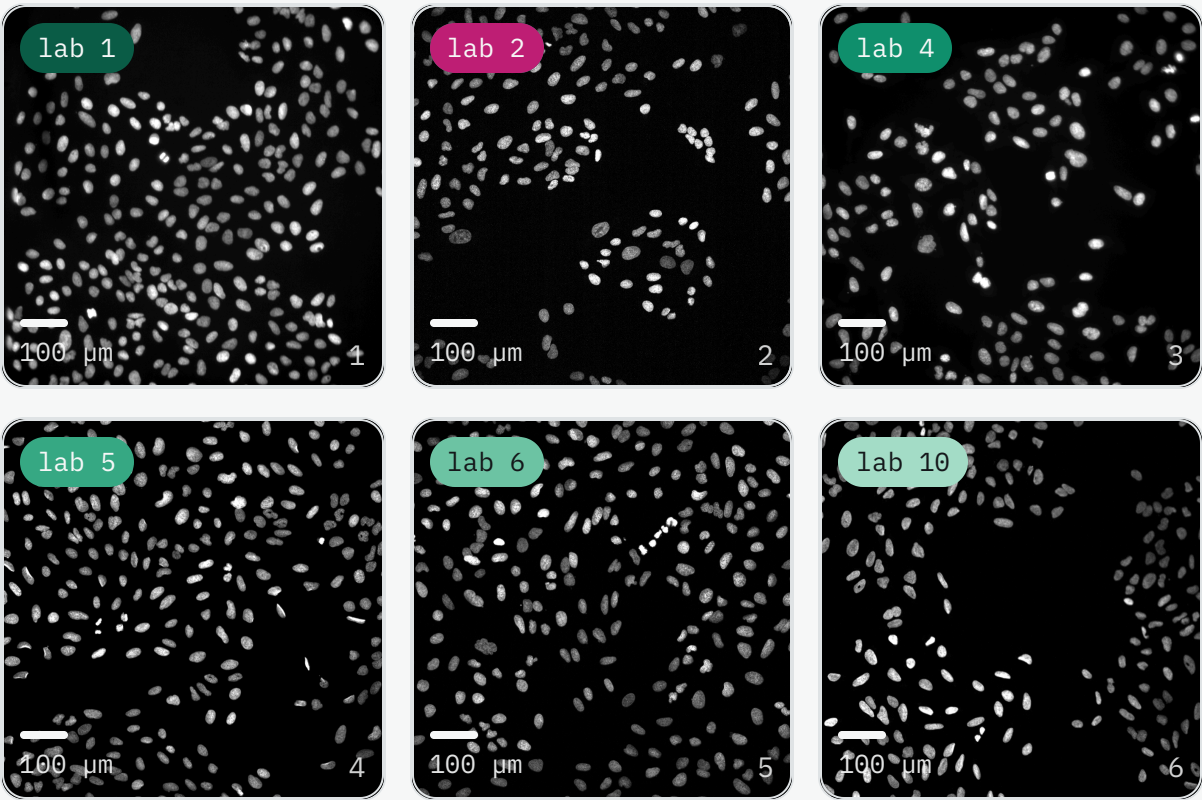
Cell Painting five dyes, 3,773 numbers per well

-  same line ✓
-  same protocol ✓
-  no drug ✓

Fig. 1 93,228 untreated wells, 11 labs, plates held out. JUMP [7]

[click → the answer](#)

Could you tell? A classifier can.



Cell Painting five dyes, 3,773 numbers per well

99.9% CLASSIFIER

9% CHANCE, 1 IN 11 LABS

78 wells wrong, of 93,228

Normalise each plate to its own controls: 99.9% becomes 24%. A3

Fig. 1 93,228 untreated wells, 11 labs, plates held out. JUMP [7]

click → follow one well

One well, **all the way out.**

ONE NUCLEUS



192 μm² · 0.59

ITS WELL

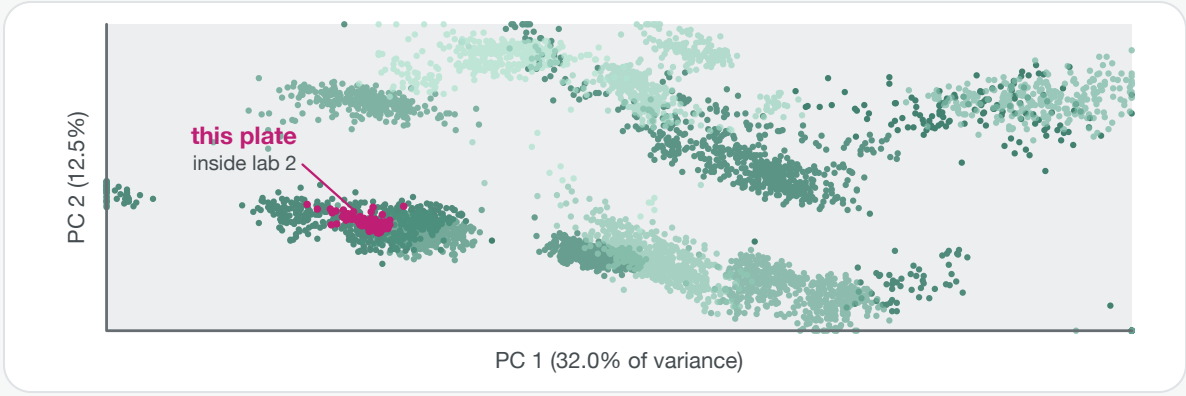


plate 1053600681

3,773
numbers
→
one point

WHERE IT LANDS

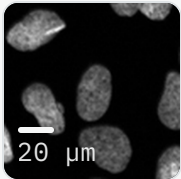
Cell Painting five dyes, 3,773 numbers per well



93,228 wells, 11 labs

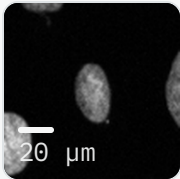
other labs

lab 5



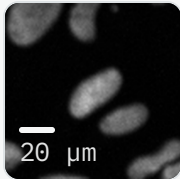
195 μm² · 0.94

lab 6



213 μm² · 0.50

lab 10



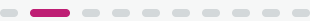
207 μm² · 0.29

AREA ≈ ≈
TEXTURE ≠ ≠

One lab's normal is not another's. The difference is shape, not count.

Fig. 1 93,228 untreated wells, 11 labs, plates held out. JUMP [7]

[click → start over](#)



Today: search, ask, guess.

WHY NOW

Since 2023 the field's reference is public [7]. Since 2025 cell data carries regulatory weight [9].

IRREPRODUCIBLE PRECLINICAL RESEARCH · US · PER YEAR [6]

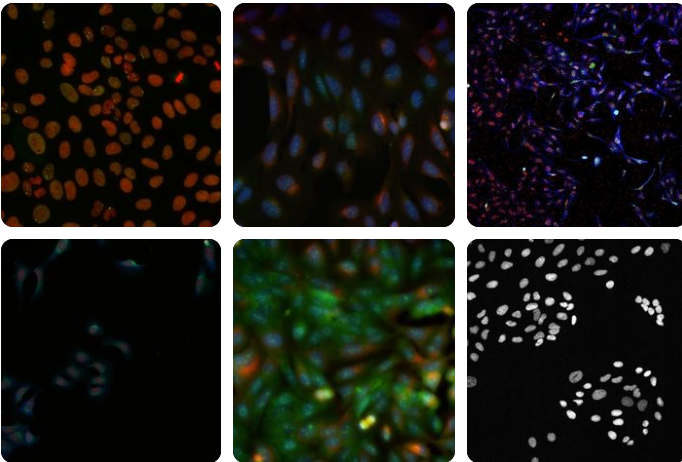
\$28B

36% ≈ \$10B · reagents & references

64% · design, analysis, protocols

1 search

Q do my U2OS cells look normal



2 ask around

To: whoever ran this line last
Subject: does this look right to you?
Attached: screenshot.png

3 guess

\$ RUN IT ANYWAY · TOO FAST

**hundreds of
thousands**

of R&D before the SOP was fixed

🕒 HOLD · TOO SLOW

months

developing cells that were already good enough

[6] Freedman et al., *PLoS Biol*, 2015. Search results: real U2OS images from six papers.



macrofluidic.me/resume

Katherine Titterton

Five labs, and the same **unanswered question** at every bench. It made the science slow and wrong.

- 2023–2026 **Axiom** SAN FRANCISCO
founding MTS • hepatocytes at scale
- 2022–2023 **insitro** SOUTH SAN FRANCISCO
neuroscience assays • productionization
- 2021–2022 **MIT** SYNTHETIC NEUROBIOLOGY
expansion microscopy
- 2018–2021 **abbvie** NEUROSCIENCE
tau • neurons
- 2017–2018  HARVARD MEDICAL SCHOOL
Xenopus oocytes • GABA_A receptors
- 2017 **COLGATE** BA
neuroscience + math biology

MY WORK

- 1 Cell Painting in primary human hepatocytes
Cell Systems 2026 • Axiom Bio
- 2 Counting cells predicts bioactivity benchmarks
Nature Communications 2026 • Axiom Bio
- 3 A neuronal tau aggregation assay
J Biol Chem 2025 • AbbVie
- 4 Multiplexed expansion revealing
Nature Communications 2024 • MIT
- 5 Neurosteroid photolabeling of GABA_A receptors
Eur J Med Chem 2019 • MGH • Harvard Medical School

PUBLISHED WITH

Broad Institute • MIT • AbbVie • EMBL-EBI • HESI • Bayer
• University of Cambridge • Uppsala University

Eleven labs. Eleven normals.

6.2

spreads between one lab's
normal and another's

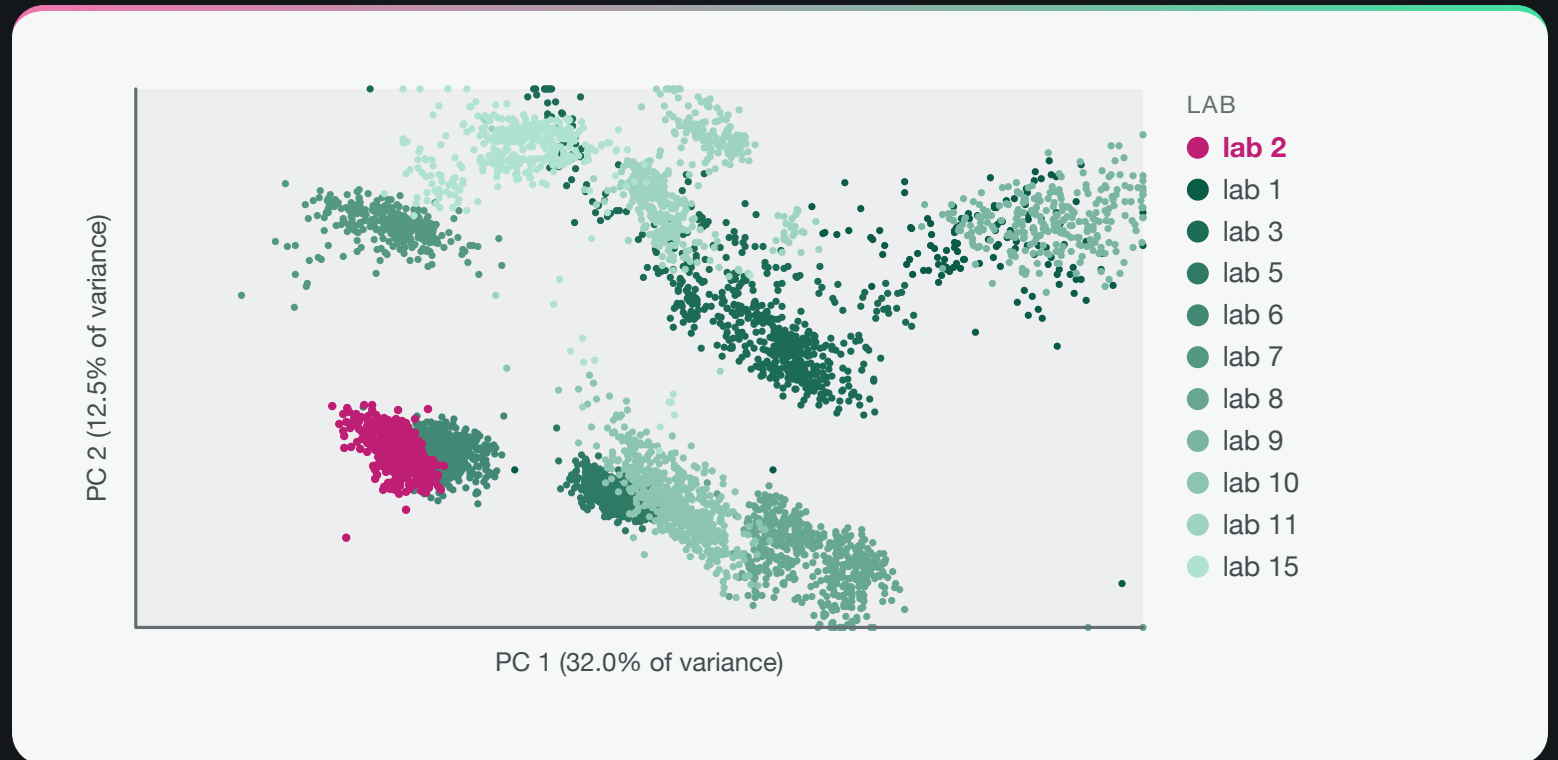


Fig. 2 15,352 untreated wells, same line, same protocol. PCA, for the eye only. JUMP [7]

[click → why the map matters](#)

Where you sit predicts what you measure.



INSIDE LABS

$$\rho = -0.71$$

the further from your baseline,
the further from your own answers

lab 2 against its own June answer

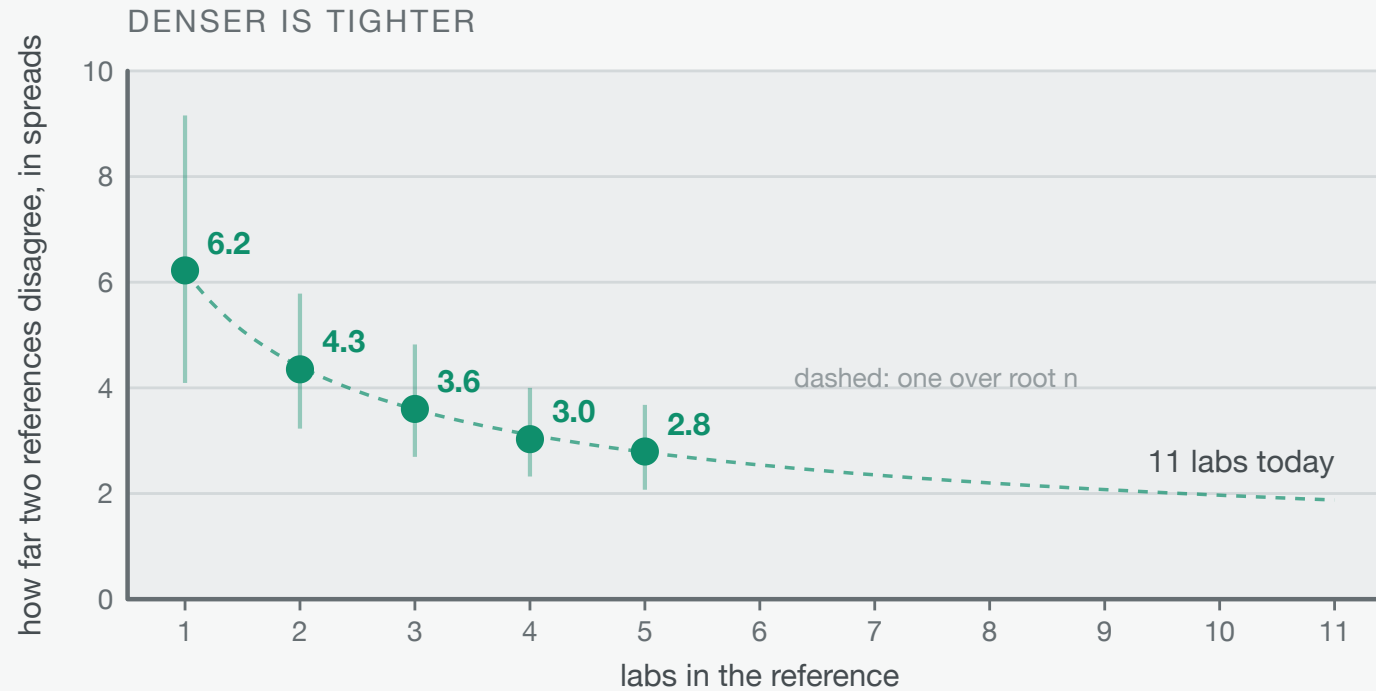
0.85 before the flag

0.67 after it

Fig. 3 One point per batch: 118 batches, 11 labs. JUMP [7]. More: A8, A13, A14

click → start over

Every lab that joins **sharpens the map.**



THE REFERENCE

6.2 → 2.8

from one lab to five:
every lab that joins sharpens
the map for everyone

your own baseline

0.9 → 0.4

from one plate to eight, then flat

COMMUNITY open

the map, the box, the code, the public reference

DATA ACCESS paid

the same data structured, comparable, pre-cleaned on request

THE BENCHMARK co-built

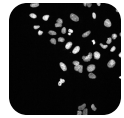
planted failures; every method graded, segmentation to bioactivity

Fig. 4 Two references, each built from the same number of labs. 1,871 plates, 11 labs. JUMP [7]. More: A14

Prices and terms proposed. A15

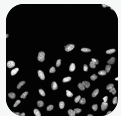
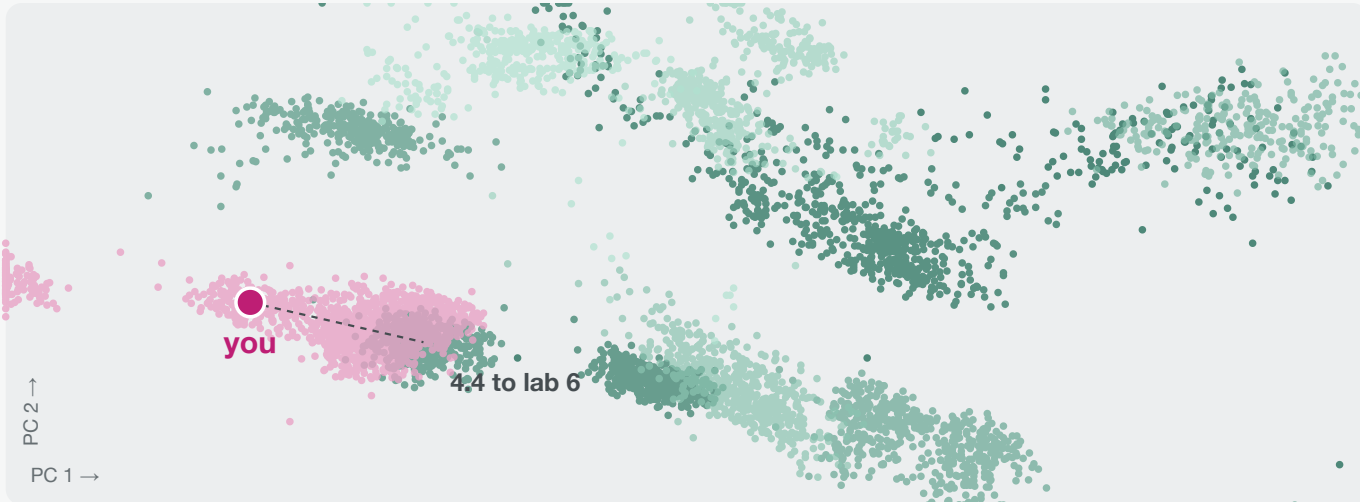
Drop an image. Get an answer.

Where your cells sit, whose they look like, and their cells next to yours.



lab 2 · plate 1086292259 · 16 August 2021
one control well, DNA channel, as imaged

answered



Yours, and lab 6's. 4.4 apart.



The answer

LAB 2 · 16 AUGUST 2021



YOUR CELLS

U2OS · 25 plates

WHERE YOU SIT

in distribution

2.8 from your baseline · 4.4 to lab 6, the nearest

YOUR OWN CHECK

188 cells per well · in range

YOUR REFERENCE

1,652 plates · 10 labs

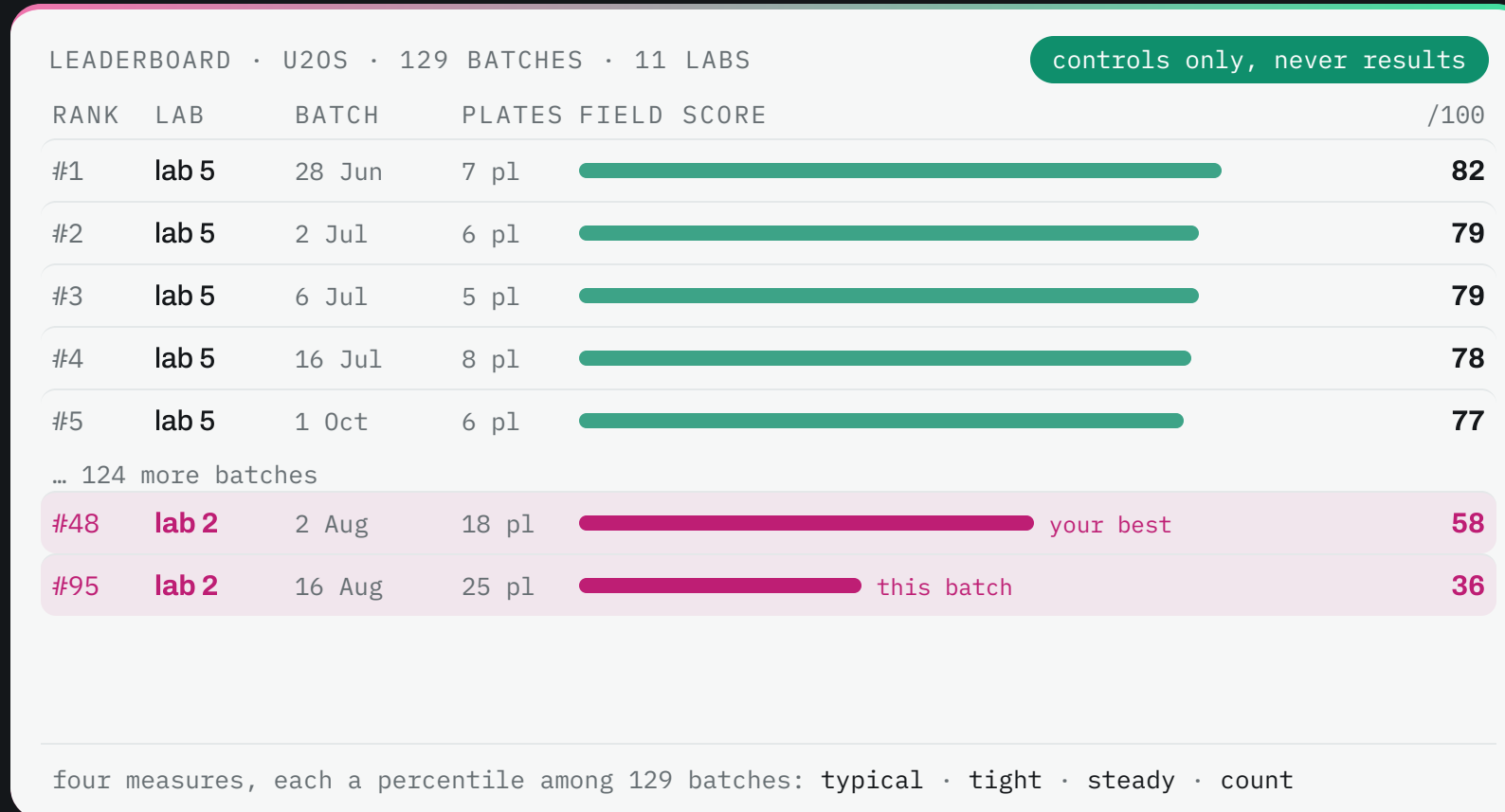
YOUR DATA, MADE COMPARABLE

agreement with other labs **0.70 → 0.77**

This page is the certificate. Held by nobody who sells you cells.

A game you can win.

Every batch ranked on its controls alone: **how typical, how tight, how steady.** Never on results.



```
click → start over
```

Your standing

LAB 2 • 16 AUG

#95 of 129 batches · score 36

YOUR BEST

#48, 2 Aug · score 58

YOUR STREAK

9 batches in a row · the field's longest: 33, lab 5

TO CLIMB

steady: 2.9 against 1.2 for the field

THE LAB TO BEAT

lab 5 · median score 70 over 33 batches

```
closest to the centre: lab 3 ·
tightest plates: lab 5 · longest
streak: lab 5
```

Five questions, five tools.

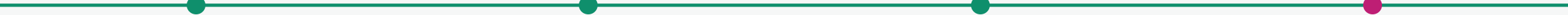
Each answers a different question about your cells. Use all of them. The last one did not exist.

TOOL	THE QUESTION IT ANSWERS	HOW	WHAT IT DOES NOT DO
STR authentication	Is it the line the label says?	a DNA fingerprint against a database; the standard for identity	nothing about how the cells behave
Vendor certificate	Was it fine when it shipped?	identity, sterility and viability at the bank	nothing after it leaves the bank
Plate controls	Did this plate work?	control wells on every plate; catches a bad well or plate	cannot see a whole lab drift, because the controls drift with it
Phenomics platforms	Is my own data consistent?	batch correction inside one organisation's data	removes lab differences by design; nothing across labs
Datum Line	Where do my cells sit, against every other lab?	one map of untreated wells from every lab; a place, a distance, a frame	cannot tell which line it is (that is STR); fluorescence today, brightfield is the bet

Partners, not rivals: an identity check and a behaviour check belong on the same certificate. More: A7, A3

Eighteen months, four milestones.

A pre-seed to run it. Each milestone is something a partner lab can see on the map.

**1**

Three labs on the map

U2OS controls from three partner labs, live certificates every batch

2

Brightfield proven

the same map from label-free images, the bet on A10

3

The benchmark, with two partners

two companies contribute failures and get their methods graded. A16

4

A first paid certificate

on a dataset a company bought and could not align

TODAY

One founder. Eleven labs of public data, one cell line, fluorescence. Every number in this deck.

Proposed. The proof of concept sets the size.

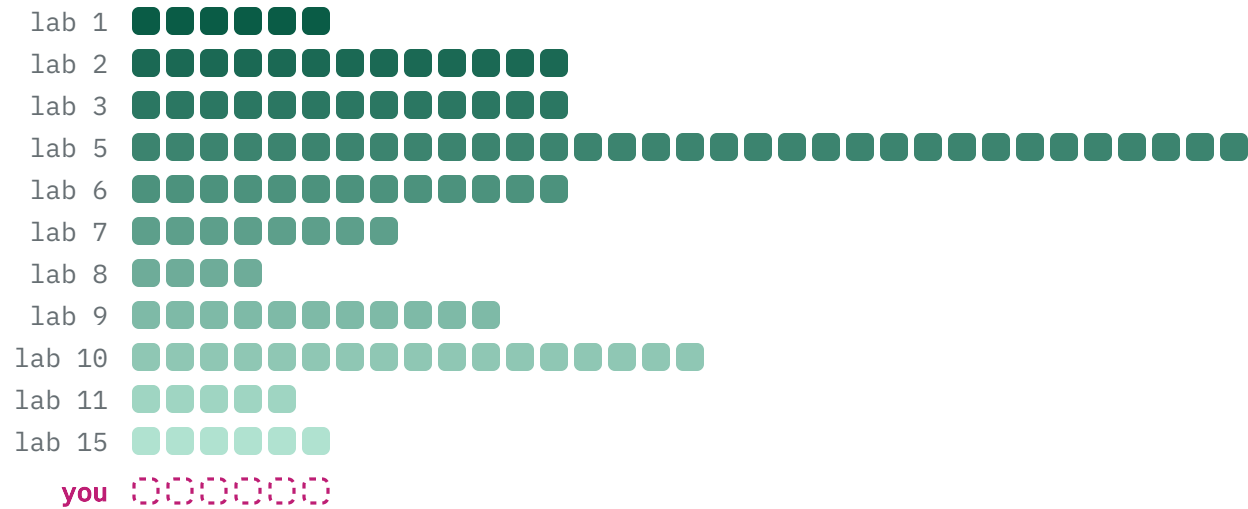
MONTH 18

→ Three partner labs live, brightfield, the benchmark with two partners, and a first paid certificate.

Every plate you image, placed on the map.

One map for people and for agents, held by nobody who sells you cells.
Everyone learns and grows together.

FIG. 7 • ALREADY ON THE MAP 129 batches • 11 labs • JUMP [7]



1,871 plates, one map. The next row is yours.

- NOW

U2OS, fluorescence
- NEXT

more lines, brightfield
- THEN

the baseline a field shares

WHAT I NEED

-  **Images** U2OS controls
-  **People** ML, normalization
-  **Bench** 3-lab proof, then the set
-  **Pre-seed** to run it
-  **Partners** two, for the benchmark

Have images of U2OS cells? Talk to me.

The rest, if you ask.

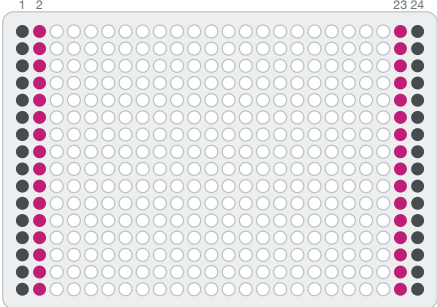
Find the question, jump to the board. Every number traces to a file in results/ or a numbered reference; pending work says pending.

A1	Why not trust the plate’s controls? There is no control well	A11	Would it cry wolf? Six of 129 batches
A2	Show me the math. Six operations on one embedding	A12	What moved? Stain intensity, not cell count
A3	Why not batch-correct it away? What JUMP records	A13	How does one lab move over time? The lab’s check and the field’s view
A4	Is the model the moat? Counting cells often wins	A14	What does the field’s frame buy? Agreement between labs
A5	Why not just count cells? Two ways to ask one question	A15	Who pays? Three buyers, one certificate
A6	How often do I re-certify? The line is not expired	A16	What would the money build? The ugliest dataset, on purpose
A7	Isn’t this STR? STR says who, images say how	A17	Is the leaderboard fair? Controls only, never results
A8	Is the assay itself working? A known answer	A18	What is private? Control wells leave; nothing else does
A9	Does it change the result? Same drug, different answer	A19	What is the technology? Geometry today; any image next; people and agents on one map
A10	Does it work without stains? Bet on brightfield		

There is no control well.

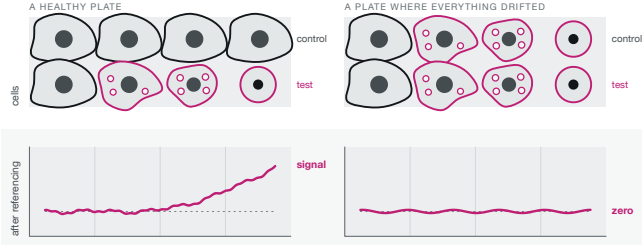
Every result is test minus control, on the same plate. But a whole plate can be off, or a whole batch, or a whole lab. A control only sees inside its own box.

A REAL JUMP PLATE · 384 WELLS



● DMSO 32 ● positive 32
○ compounds 320

The pink wells are the plate's zero: 32 of 384, near the edge by design.



A control only sees inside its own box.

Whole plate off: its controls move with it.
Whole batch off: so do every plate's.
Whole lab off: nobody inside it can tell.
Only the field sits outside every box.

$$x_{test} = \mu + s + \tau(s) \quad x_{ctrl} = \mu + s \quad \Delta = \tau(s)$$

s , the plate's state, cancels from the number. Never from the cells.

So a control only sees inside its own box. If the plate is sick, so is the control. Then how would anyone know?

Fig. A1 Where a control sits, and what it can see. Left, a real 384-well JUMP compound plate (lab 2, 1053601756): 32 DMSO, 32 positive, 320 compound wells. Right, schematic cells and traces, and the nesting of well, plate, batch, lab and field. No measurements shown.

Six operations on one embedding.

Every well becomes a vector $e \in \mathbb{R}^{3773}$, z-scored across the field. Everything on the previous slides is one of these, and each one is a difference of means or a held-out classifier, so a lab can check it.

1 · EMBEDDING

[1]

$$e = \varphi(\text{image}); \varphi \in \{\text{CellProfiler}, \text{CNN}, \text{ViT}\}$$

Which φ barely matters: the three performed similarly on hepatocytes. The extractor is not the asset.

2 · LAB CLASSIFIER

99.9%

$$f: e \rightarrow \text{lab}, \text{ held out by plate}$$

Logistic regression, L2, five folds. Balanced accuracy, chance 1/11. Permuted labels 9.3%. Without the 1536-well labs: 100.0%.

3 · DISTANCE, THE CERTIFICATE

[e6]

$$d = \| E[e \mid \text{plate}] - E[e \mid \text{field}, c] \|_{\sigma}$$

Difference of two means in units of the field's spread, conditioned on c . Its direction is what moved; $-d$ is the way back.

4 · VARIANCE DECOMPOSITION

[e5]

$$\text{Var}(e) = V_{\text{model}} + V_{\text{lab}} + V_{\text{batch}} + V_{\text{plate}} + V_{\text{well}}$$

Nested, so shares sum to one. Microscope model separates from lab because models are shared across labs.

5 · REGRESS VARIABLES OUT

24%

$$e_{\perp} = e - \hat{E}[e \mid c], \text{ then re-run } f$$

Remove only what you know is instrument, keep what may be biology. Remove each plate's own control mean and f falls to 24%.

6 · MORPHOLOGY CHANGE, TWO WAYS

 δ

$$\text{DISTANCE } \delta = f(\|e - E[e \mid \text{vehicle}]\|), f \text{ bounded}$$

$$\text{CLASSIFIER } g: e' \rightarrow \text{label}, e' \text{ aligned to the reference before comparison}$$

Same vector, two questions. Discovery wants invariance to the lab: align first. QC wants the lab coordinate: never align it away.

results/e1, e1b, e5, e6 · [1] Ewald et al. 2026

What JUMP records. What nobody does.

Normal is a conditional expectation, $E[x \text{ / every variable below}]$. Even JUMP, built to be comparable, records only the instrument side.

RECORDED IN JUMP

- lab
- batch date
- plate
- plate type
- well position
- microscope
- confocal or widefield
- laser or LED
- objective NA
- fields per well
- filter set
- pixel size
- brightfield planes
- CellProfiler version

NOT IN ITS PUBLIC METADATA

- DMSO % per well
- DMSO lot
- DMSO water uptake
- stock freeze-thaw
- serum lot
- media lot
- passage
- thaw date
- seeding density
- confluence at dosing
- dosing to fixation
- stain and dye lots
- operator
- incubator shelf
- edge evaporation
- mycoplasma status
- CO2 and humidity

NORMALISE THE LAB AWAY AND THE LAB DISAPPEARS

- 99.9% raw features
balanced accuracy naming the lab
 - 24% plate-centred
own control mean removed; chance 9%
 - 19% plate-centred and scaled
own control spread removed too
- 29,943 wells, 16 per plate, plates held out. results/e1-redteam.

Centre each plate on its own controls, the standard first step, and the lab is named 24% of the time. It removes most of the coordinate a certificate needs.

Batch correction is built to remove this signal: Arevalo J, et al., Nature Communications, 2024. Variables: JUMP datasets metadata.

And the benchmarks are so weak that **counting cells** often wins.

If a naive baseline matches your model, the benchmark is the problem. The bottleneck was never the model. It is what you are comparing against.

Two ways to ask the same question.

Why now: an image classifier can answer this, and until recently nothing could. A count moves only when cells die or divide; an embedding of the whole well moves when anything changes.

SAME CELL LINE, SAME PROTOCOL



cell count	236	224	same
shape	0.94	0.50	different

METHOD ONE

Does this run belong with the past runs?

$$g: e' \rightarrow \text{belongs} / \text{does not}, \quad e' = \text{align}(e)$$

Align first, so the answer is the cells and not the day.

Standard alignment; the tuning is ours.

METHOD TWO

How far out, and which way back?

$$\delta = f(\|e - E[e \mid \text{vehicle}]\|), \quad f \text{ bounded}$$

Bounded, so one wild well cannot swamp the score.

Distance published with the Broad [1]; transform unpublished.

Both methods need the same thing to work: somebody else’s images. It takes a reference to know which.

Fig. A5 Same cell line, two untreated wells, and the two readouts. Left, DNA channel, JUMP cpg0016, CC0; nuclei by Otsu segmentation; detail = high-frequency energy over total, inside each nucleus. Right, the two readouts: a classifier on aligned embeddings, and a bounded distance from the vehicle centre. Alignment and distance follow published practice [1]; the specific transform and tuning are not disclosed here. [1] Ewald JD, Titterton KL, et al. *Cell Systems*, 2026.

The line is not expired. Your baseline is.

Two plates of one lab sit further apart the longer the gap between them. In lab 2 the distance is still rising after thirteen weeks.



IS THE CELL LINE RUINED?

Not known from profiles alone. The data says the lab moved and when. It does not say why.

SO WHY RE-CERTIFY?

Because your certificate is a comparison made on one day, and the baseline it was made against goes stale within weeks.

HOW SOLID IS THE NUMBER?

One lab, 219 plates, 13 batches, June to October 2021. 0.63 in the same batch, 4.7 over 13 weeks. Other labs are not yet measured.

STR says who.

Images say how.

Not a competitor: the other half of the same certificate. Every lab already runs the left column. The right column costs nothing extra.

	STR PROFILING	CONTROL-WELL IMAGES
WHAT IT READS	DNA at 17 short tandem repeat loci	the pictures the screen already took
WHAT IT PROVES	the line is what the label says	the cells still behave like the field's version of that line
WHAT IT MISSES	two cultures, same fingerprint, different behaviour	identity: it cannot tell you the line is mislabelled
SAMPLE	cells spotted on a card and mailed	nothing leaves the building but images
EXTRA WORK	a separate prep, per line, per test	none: control wells you already image
CADENCE	on receipt, after 10 passages, when in doubt	every quarter, from the last plate you ran
ANSWER	a profile to compare by eye or by database	one of two words, and a distance

Identity once. Behaviour every plate. One line on the certificate each.

STR practice and cadence: ATCC cell authentication service and test recommendations; 17 loci plus amelogenin on the Promega PowerPlex 18D system. Low- and high-passage LNCaP respond differently to the same hormones: ATCC technical document.

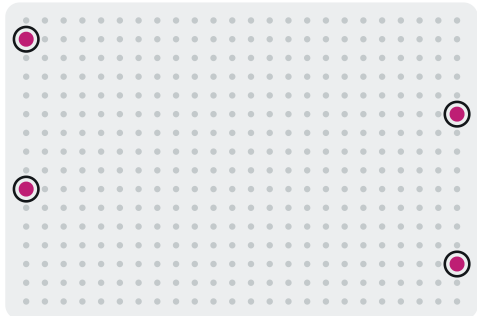
A known answer, run in every lab.

Untreated wells certify where a lab sits. Positive controls certify that the assay answers at all. The field already runs the same eight.



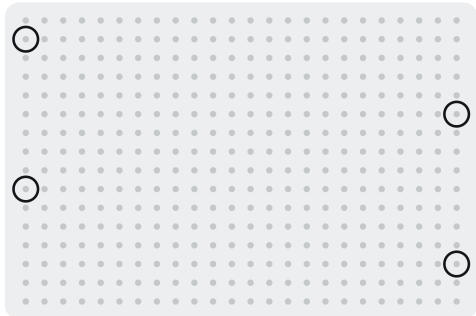
WHERE ON THE PLATE DOES AMG900 SHOW?

a batch that came out
plate C13443aW



○ where the plate map puts it

a batch that failed
plate A12069aW



● wells that look like it

RUNS NOW ON JUMP

Every compound plate carries eight positive controls. In 125 of 129 batches the known answer came out.

WHAT THE FIELD CAUGHT

Lab 3, 4 consecutive batches, 65 plates: the wells the plate map labels as positive controls show no effect, and no other well on the plate does.

TWO CHECKS, TWO FAILURES

The untreated-well certificate flagged 1 of these 4 batches. A plate can sit in the right place and still not answer.

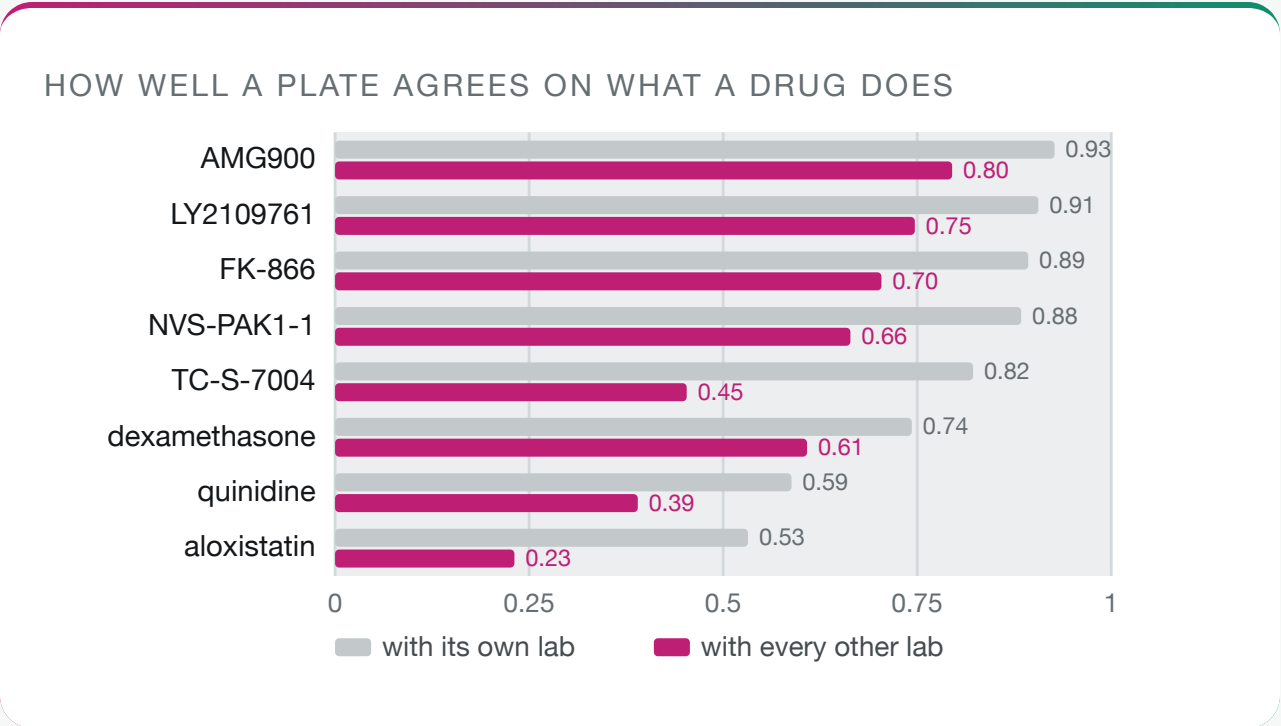
Same drug. Different answer.

The field already contains the experiment: eight fixed compounds, each measured 1,829 times, in 11 labs.

WHAT A SCREEN REPORTS, PER PLATE

$$\delta_{p,c} = E[x \mid \text{plate } p, \text{ compound } c] - E[x \mid \text{plate } p, \text{ DMSO}]$$

Treated minus control, on the same plate. Scaling fit on the field's DMSO wells only, never on the plate being judged.



Q1 · DOES IT AGREE? [E8]

Less with other labs, for every compound.
AMG900: 0.93 with its own lab, 0.80 with the others.

Q2 · DOES IT TRACK THE CONTROLS? [E8]

Inside labs, yes.
Spearman 0.20 to 0.53 by compound; pooled 0.30, shuffled 0.01. Between labs: same sign, too few labs.

Q3 · DOES IT MATTER? [E8]

Effect size differs 14% to 79% between labs.
Inside one lab: 10% to 44%.

Shown in fluorescence.

Bet on brightfield.

Every number here is Cell Painting: five stains. The product reads brightfield: none. That gap is the bet, testable on public data.

WHAT IS SHOWN

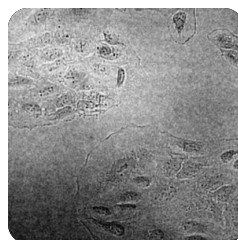
Stained wells name the lab

Untreated U2OS wells, 11 labs: the lab is named 99.9% of the time, chance 9%.
results/e1

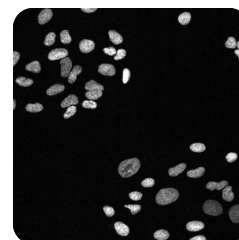
WHY THE BET IS FAIR

Fewer variables, not more

No stain lot, dye lot or staining step:
three unrecorded variables gone (A3).
JUMP images brightfield beside the
stains, so the same wells re-run without
them.



BRIGHTFIELD



DNA STAIN

One field of
one well,
two ways.
Lab 2, June
2021.

WHAT WOULD KILL IT

Brightfield alone at chance

Then the certificate asks for one cheap
stain. The test: e1 on the brightfield
planes of the same 93,228 wells.
[pending]

Brightfield timecourse is the modality I build screens on.

Would it cry wolf? 6 of 129.

A flag is only worth something if it is rare, and if it survives the choices made to compute it. Both are measured.

LAB 2 · THE SAME 13 BATCHES, JUDGED TEN WAYS

	7	14	21	12	19	26	2	8	16	23	30	20	3	
	Jun	Jun	Jun	Jul	Jul	Jul	Aug	Aug	Aug	Aug	Aug	Sep	Oct	
as on slide 06	■	■	■	■	■	■	■	■	■	■	■	■	■	3,773 features
baseline is the first two batches	■	■	■	■	■	■	■	■	■	■	■	■	■	3,773 features
baseline is the first three batches	■	■	■	■	■	■	■	■	■	■	■	■	■	3,773 features
no clipping	■	■	■	■	■	■	■	■	■	■	■	■	■	3,773 features
no image-level features	■	■	■	■	■	■	■	■	■	■	■	■	■	2,904 features
image-level features only	■	■	■	■	■	■	■	■	■	■	■	■	■	869 features
all-vehicle plates removed	■	■	■	■	■	■	■	■	■	■	■	■	■	3,773 features
lab 1 removed from the field	■	■	■	■	■	■	■	■	■	■	■	■	■	3,773 features
lab 8 removed from the field	■	■	■	■	■	■	■	■	■	■	■	■	■	3,773 features
labs 1 and 8 both removed	■	■	■	■	■	■	■	■	■	■	■	■	■	3,773 features

HOW RARE

6 of 129 batches are outside, in 3 of 11 labs. 8 labs never flag.

DOES EVERY PLATE AGREE

In lab 2's flagged batches 25 of 25 plates are outside. In its other 10 batches, 2 of 194.

WHAT CHANGES IT

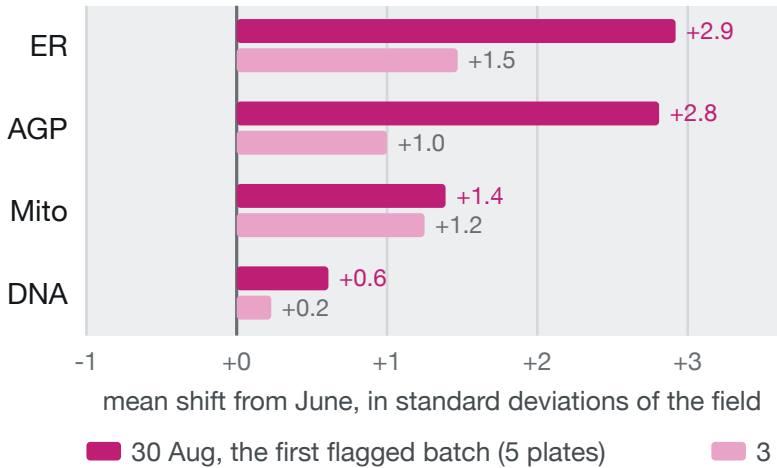
1 of 10 variants: a baseline that takes in 21 June, the batch whose cell count fell. A baseline should be a batch the lab trusts.

Outside: further from its own first batch than from the nearest other lab. Flagged batches hold 5, 6, 14 plates.

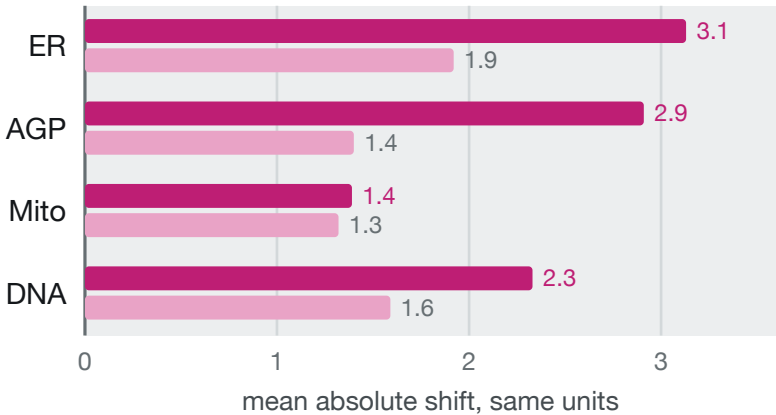
Stain intensity rose. Cell count did not.

The certificate's distance is a difference of two means, so it reads feature by feature.

STAIN INTENSITY



TEXTURE



READING

On 30 Aug, ER and AGP intensity sit 2.9 and 2.8 sd above June, and 96% or more of those features moved the same way. By 3 Oct: 1.5 and 1.0. Cells per well: 189, then 165 and 182.

INSTRUMENT OR CELLS

Image-level features carry 20% of the move and are 23% of the features, so the move is not confined to acquisition. Profiles cannot tell a stain lot from biology.

WHAT A LAB GETS

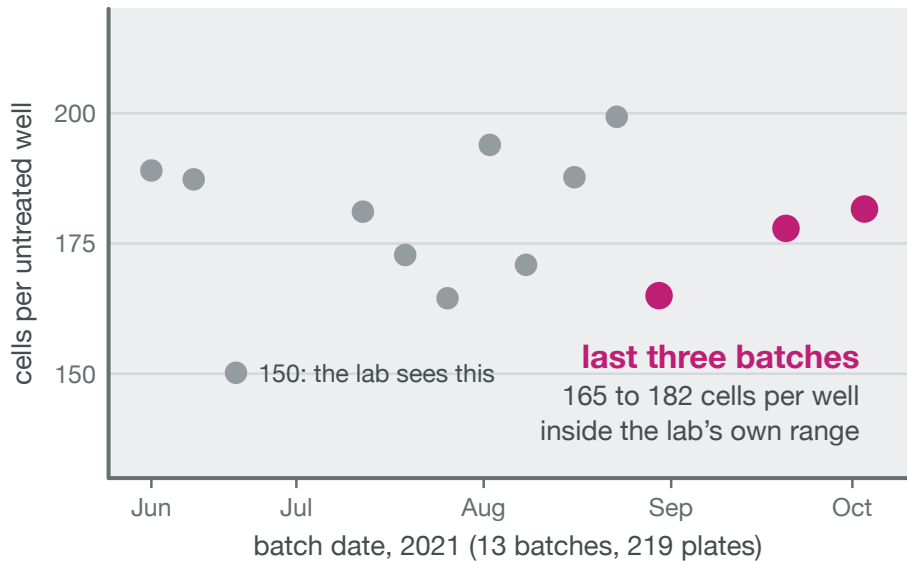
Where to look first: staining and exposure in two channels. Not why. The cause is the lab's to find, a day after the run and not four months.

Shift from the lab's first batch, in standard deviations of the field; means over the features of one channel.

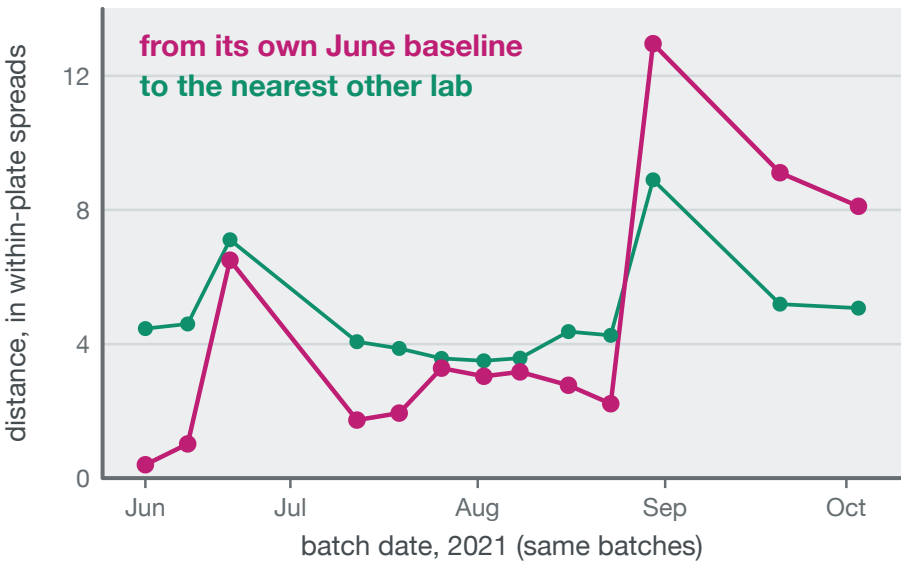
One lab, 13 batches.

The lab’s own check and the field’s view of the same batches, side by side.

WHAT THE LAB CAN SEE



WHAT ONLY EVERYONE ELSE CAN SEE



THE LAB’S OWN CHECK

Cells per untreated well stay between 150 and 199. The last 3 batches sit inside the range of the earlier ones.

THE FIELD’S VIEW

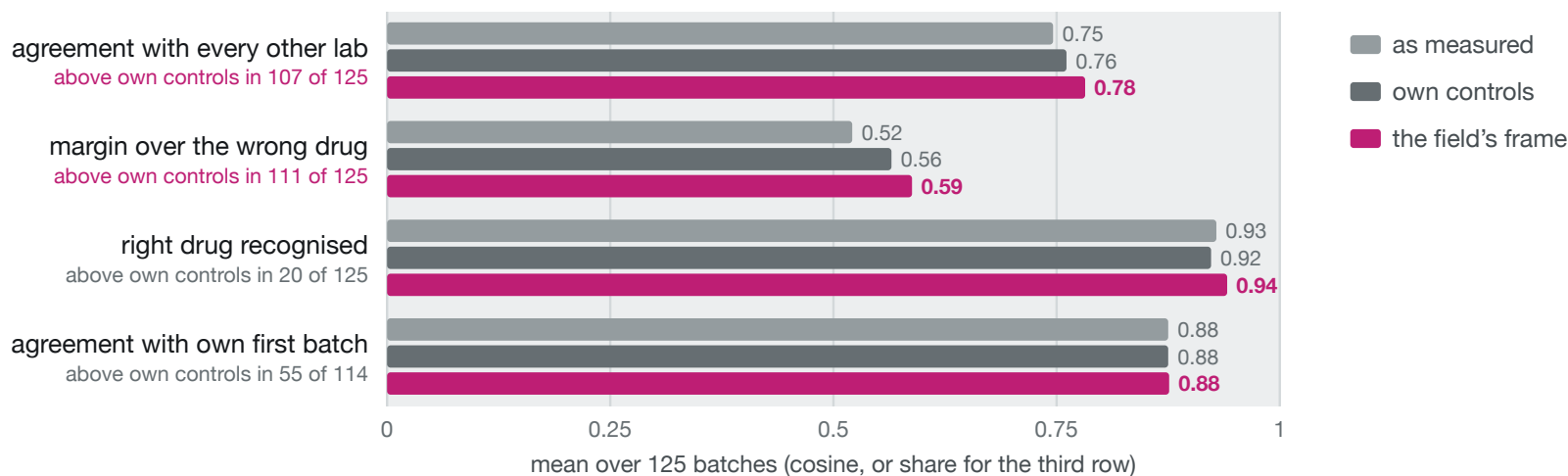
From 30 Aug the lab is further from its own June baseline than from another lab. 3 Oct: 8.1 from baseline, 5.1 to lab 8.

WHY BOTH

Neither replaces the other. The count is fast and local. The field’s view needs everyone else’s images.

One frame: **what it buys.**

Each batch's untreated wells are moved onto the field's, built without that lab. The drugs are the test.



DOES IT HOLD

With 100, 200 or 400 components and shrinkage from 0.05 to 0.5: above own controls in 89 to 116 of 125 batches for agreement, 52 to 123 for margin.

WHAT IT DOES NOT DO

A lab's agreement with its own first batch does not change: 0.88, then 0.88. Effect sizes vary more between batches: 0.06 becomes 0.12. The gain is between labs, and it is modest: 0.76 to 0.78 over own controls.

HOW DENSE

A lab's own baseline settles by eight plates: 0.89 to 0.39 spreads. The field's reference moves 0.41 when one lab leaves.

Three buyers, one certificate.

Contributors use it free. The labs that cannot afford to be wrong pay.

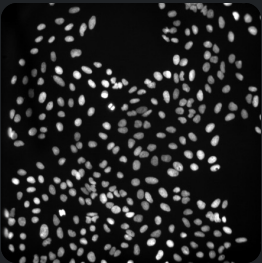
	 ACADEMIC LABS AND CORES	 PHARMA AND CRO SCREENING	 CELL BANKS AND VENDORS
What is at stake	the paper	the screen · ~\$2.6M	the lot
What they pay	free, for contributing controls	~\$5–15k per line, per quarter 170× less than one unchecked screen	a fee per certified lot
What it builds	the reference	the revenue	the network

Build the ugliest dataset in biology, **on purpose.**

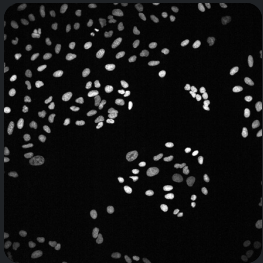
Same cells, same protocol, no drug. It already looks like this.

THE TEST SET · WHAT SIX LABS ALREADY SEND

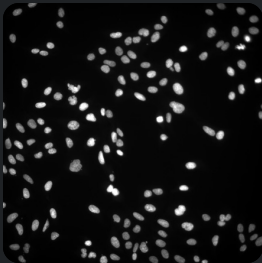
real wells, JUMP cpg0016, CC0



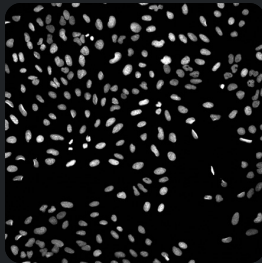
LAB 1
Phenix · widefield



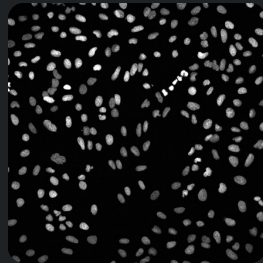
LAB 2
CV8000 · confocal



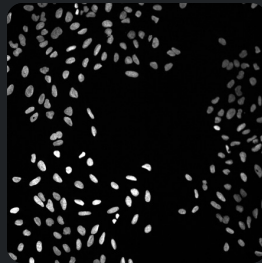
LAB 4
Phenix · widefield



LAB 5
CV8000 · confocal



LAB 6
CV8000 · confocal



LAB 10
CV8000 · confocal

5
microscope models

0.4–1.0
objective NA

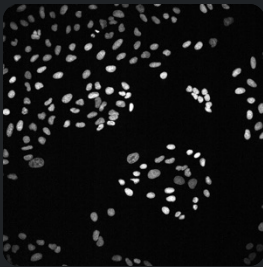
0–3
brightfield planes

0.60–0.70
µm per pixel

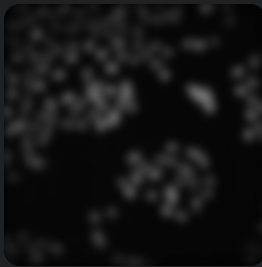
8
plate types

Across the 13 JUMP labs.

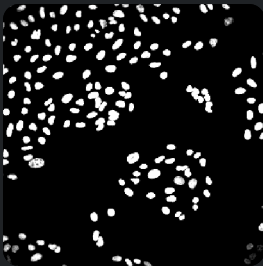
RED TEAM one well, simulated



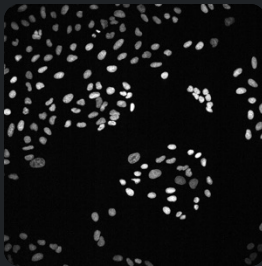
AS IMAGED



DEFOCUSED



OVER-EXPOSED



CLIPPED WELL

A public benchmark: name the failure.

The benchmark every method is graded on, normalization to bioactivity: the glue between the labs that make data and the companies that b

The leaderboard.

Controls only.

Every batch of every lab, ranked on its untreated wells. A lab can climb only by growing and imaging cells more like the field, more alike and more steadily.

LAB	BATCHES	MEDIAN SCORE	BEST	STREAK	MEDIAN TYPICAL	MEDIAN TIGHT	MEDIAN STEADY	MEDIAN COUNT
lab 5	33	70	#1	33	3.35	0.27	0.81	0.31
lab 6	13	60	#16	13	4.44	0.41	1.23	0.12
lab 10	17	54	#10	17	3.84	0.62	1.21	0.23
lab 8	4	52	#54	4	4.10	1.08	0.86	0.09
lab 11	5	47	#23	2	3.79	0.87	1.57	0.17
lab 3	13	44	#18	6	3.27	0.80	2.01	0.29
lab 7	8	42	#69	8	6.38	0.58	1.17	0.33
lab 2	13	39	#48	10	5.68	0.57	1.99	0.14
lab 1	6	32	#78	6	8.68	1.10	2.03	0.09
lab 9	11	25	#97	11	9.97	0.50	1.76	0.45
lab 15	6	23	#68	6	5.53	0.87	1.69	0.44

THE SCORE

Four measures in within-plate spreads, each a percentile among 129 batches: distance from the field’s centre, spread of the batch’s plates, distance from the previous batch, cells per well against the field. 100 minus the mean.

THE BEST BATCH

lab 5, 28 Jun, 7 plates: score 82. Lab 5 holds 9 of the top ten and the longest streak, 33.

WHY IT IS SAFE

Nothing in the score touches a treated well or a result. The only way up is better culture and imaging practice. The judged lab never sets its own scale.

Control wells leave. Nothing else does.

A control well has no compound, no target and no result, so there is nothing to clear. The map is built from those wells alone; the frame it produces is applied to your treated data by you.

WHAT LEAVES THE BUILDING

Untreated control wells only

images or the well-level profile of the wells with no compound. No treated well, no compound identity, no target, no result.

A plate map of the control wells

which wells are controls; nothing about what the other wells hold.

Nothing else

on this public replay that is 82,319 untreated wells from 10 other labs, and not one treated well.

WHAT DATUM LINE DOES WITH IT

Builds the map

one centroid per plate, in a space scaled on every other lab. The lab being placed never sets its own scale (3,773 features here).

Answers

where the plate sits, distance from the lab's own baseline and to the nearest lab, cells per well, the verdict, the rank.

Computes a move, not a model of you

the field's frame is one matrix per batch, fitted on your controls: $x' = C_{\text{field}}^{\frac{1}{2}} C_{\text{batch}}^{-\frac{1}{2}} (x - m_{\text{batch}})$, 200 components.

WHAT IT NEVER SEES

Your treated data

the move is sent to you and applied on your side. Your results are normalised in your building, by you.

Your name, unless you choose

labs appear on the map and the board as a number by default; a lab opts in to a name. Every JUMP lab here is a number.

Your images, for long

proposed: images are reduced to profiles on arrival and the originals deleted; the profile of a control well is what is kept.

Every analysis in this deck runs on this rule and is public: github.com/kat-titter/datum-line/analysis. The public map is JUMP, CC0 [7]. Retention and naming are proposed, not yet contracted.

Geometry today. Any image next.

The hard part was never the model. It is the reference, and the rule that nobody sets their own scale. Humans and agents use the same map, for different reasons.

TODAY: GEOMETRY, NOT A MODEL

Profiles in, distances out

3,773 CellProfiler numbers per well. Z-scored on the other labs, clipped, one centroid per plate. The lab being judged never sets its own scale.

Three small pieces of maths

a logistic regression names the lab (99.9%); a distance in within-plate spreads places a batch; one matrix per batch, 200 components, moves it into the field's frame.

A laptop

the whole replay, 1,871 plates and 11 labs, runs from cache in minutes. No GPU in this deck. Every script is public.

NEXT: ANY IMAGE

The box takes a picture

a frozen vision model, no training on cells, asked to name the lab from one raw image and to agree with the profile map. Running; results/e15 when done.

Then a model of cells

a vision model trained on the open map and graded on the benchmark; brightfield first, because that is the bet (A10).

Treated data trains in place

the map is built from control wells, which can travel. Treated wells never do: the frame is sent to the lab, and later, models learn from treated data where it sits, without moving it. Align first; only then can learning cross labs.

WHO USES IT, AND WHY

People

the box, the glimpse of everyone else's cells, the game. An answer in a second, and a reason to come back next batch.

Agents

the API: structured, comparable data with provenance, a coordinate system to reason in, and a benchmark to be graded on. An agent that cannot tell lab from biology cannot learn biology.

Both, together

every drop sharpens the map people read and agents train on; every graded method raises the benchmark both are held to. Everyone learns, and the map grows with them.